

Cars, 5G and Patent Fees

Who makes the world's cars, and what could carmakers pay for 4G/5G patents?

A beginner's market snapshot in Excel | Tanvir | October 2026

Cars are becoming connected devices, and each one carries a patent fee

- Most new cars now have a built-in mobile connection (4G or 5G) for maps, safety calls and software updates.
- That technology is protected by patents owned by telecom companies such as Nokia.
- Carmakers can license these patents through a patent pool (Avanci), which publishes a fixed fee per car.
- **My question: how many cars are made, where, and how big could these fees be?**

96.4 m

cars produced worldwide in 2025

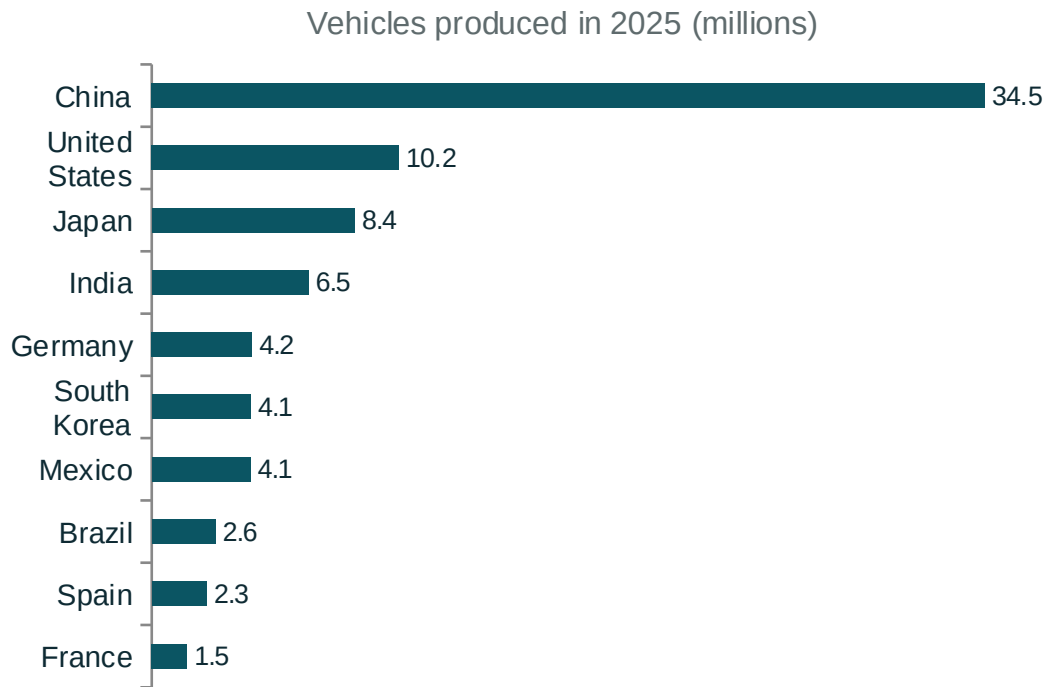
75%

of new cars have a built-in mobile connection

\$20 → \$32

patent fee per car: 4G vs. 5G

10 countries make 81% of the world's cars, and China alone makes 36%



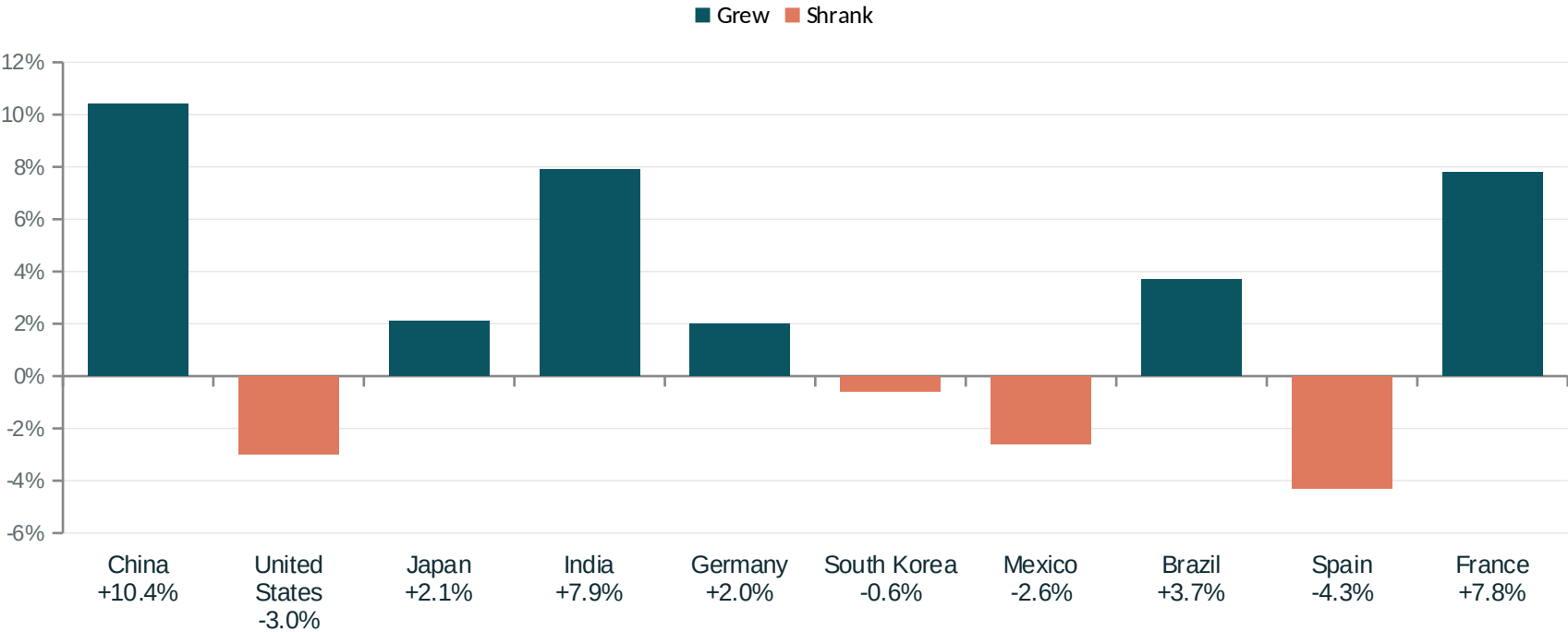
81%

of world production comes from these 10 countries

3.4x

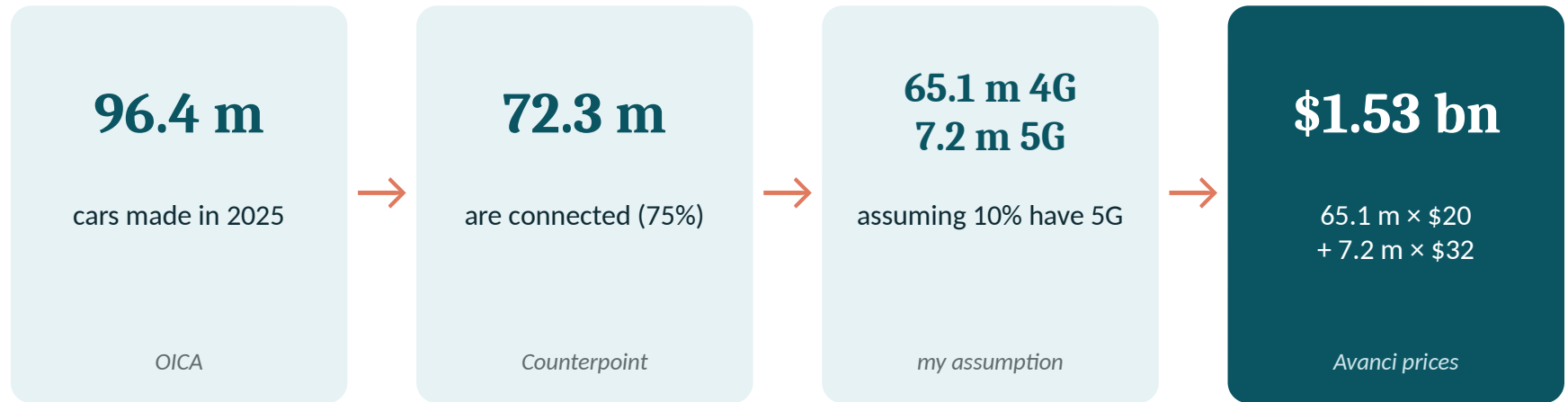
China makes 3.4 times as many cars as the US, the no. 2

China grew 10% in 2025, while 4 of the top 10, including the US, made fewer cars



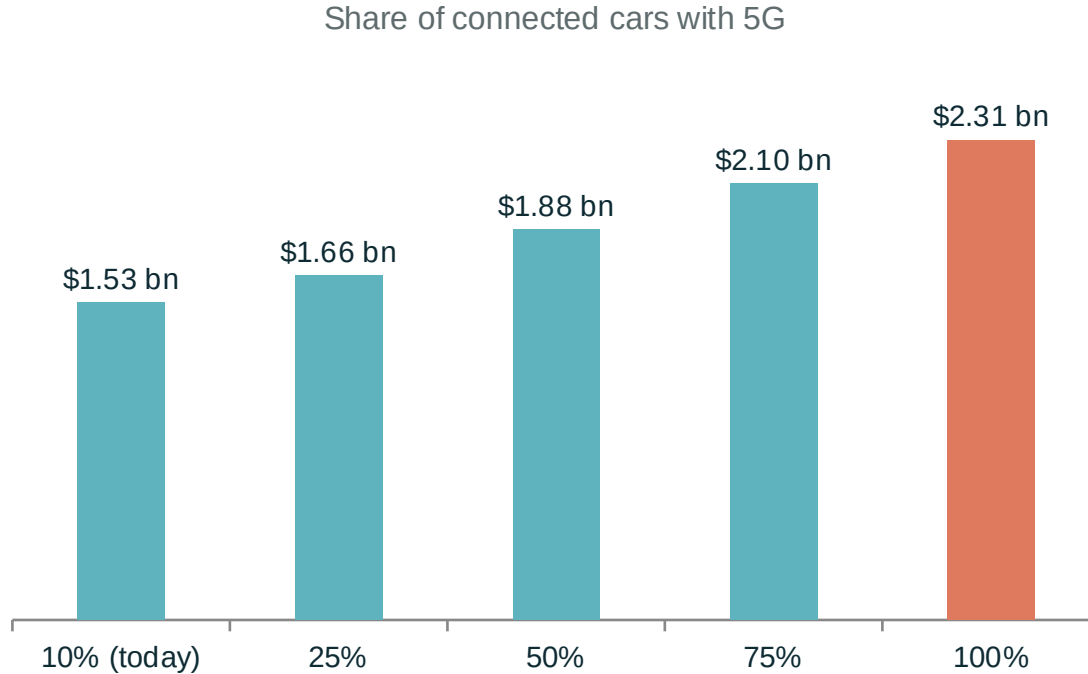
Growth = (2025 - 2024) / 2024. World production grew 4.2%; the top 10 grew 4.8%. Source: OICA.

At public pool prices, carmakers would pay about \$1.5 billion a year in 4G/5G patent fees



This is an illustrative estimate using public patent-pool prices, not any company's actual licence deals.

If every connected car had 5G, patent fees would rise by about 50% to \$2.3 billion



Why it grows

- A 5G car pays \$32 instead of \$20, which is 60% more per car.
- So fees can grow even if car sales stay flat.
- The shift from 4G to 5G is the main growth driver for patent owners.

Key takeaways

1

Car production is concentrated

10 countries make 81% of cars; China alone makes 36% and is still growing fast.

2

Patent fees are already a \$1.5 bn market

That is at public pool prices, with most cars still on 4G.

3

5G is the growth driver

Moving every connected car to 5G would lift fees by about 50%.

Skills used

- Excel: SUM, AVERAGE, MIN/MAX, RANK, COUNTIF, INDEX/MATCH
- Conditional formatting and charts
- Simple what-if analysis
- PowerPoint storytelling

Next step: compare individual carmakers to see which are growing fastest in Europe.